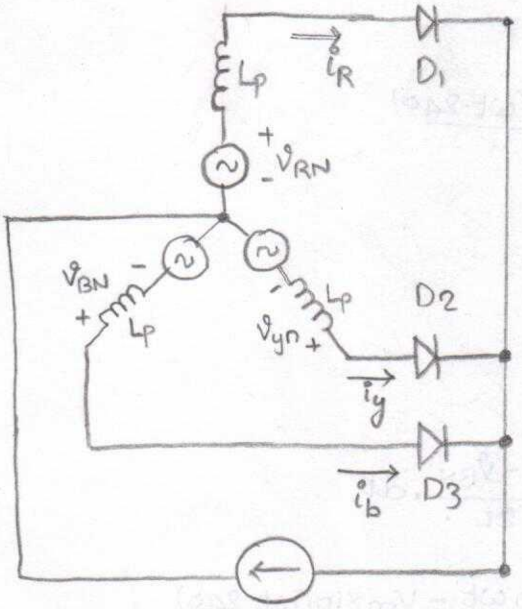
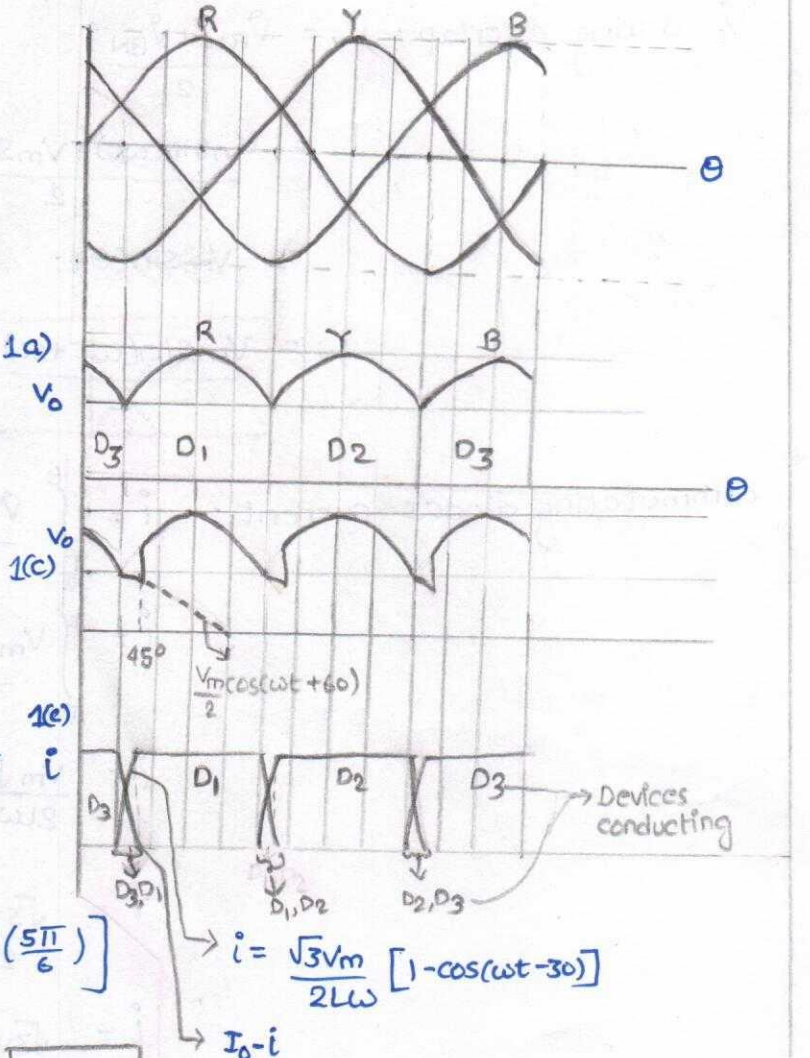
Tutorial-5

Three phase Rectifiers solutions

1



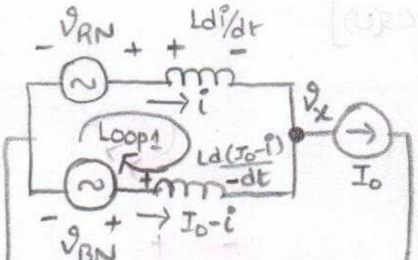


1b) $$\langle V_o \rangle = \frac{3}{2\pi}\int_{\pi/6}^{5\pi/6} V_m \sin(\omega t)\, d\omega t$$

$$= \frac{3V_m}{2\pi}\left[\cos(\pi/6) - \cos\left(\frac{5\pi}{6}\right)\right]$$

$$= \frac{3V_m}{2\pi}\left[\frac{\sqrt{3}}{2} - \left(-\frac{\sqrt{3}}{2}\right)\right] = \boxed{\frac{3\sqrt{3}V_m}{2\pi}}$$

1c) $L_p \neq 0$ @ $\theta = 30°$ (current commutation between R & B phases)



KVL in loop 1:

$$-v_{RN} + L\frac{di}{dt} - L\frac{d(I_o - i)}{dt} + v_{BN} = 0$$

$$\boxed{L\frac{di}{dt} = \frac{v_{RN} - v_{BN}}{2}}$$

$$v_x = v_{RN} - L\frac{di}{dt}$$

$$\boxed{v_x = \frac{v_{RN} + v_{BN}}{2}}$$

v_0 during overlap: $v_x = \dfrac{v_{RN} + v_{BN}}{2}$

$$= \frac{V_m \sin(\omega t) + V_m \sin(\omega t - 240)}{2}$$

$$= \cancel{V_m \sin(\theta +}$$

$$= \boxed{\frac{V_m \sin(\omega t + 60)}{2}}$$

commutating diode's current: $i = \displaystyle\int_{T/12}^{t} \frac{v_{RN} - v_{BN}}{2L} \, dt$

$$i = \int_{T/12}^{t} \frac{V_m \sin \omega t - V_m \sin(\omega t - 240)}{2L} \, dt$$

$$i = \frac{V_m \sqrt{3}}{2L\omega} \int_{30^\circ}^{\omega t} \sin(\omega t - 30) \, d\omega t$$

$$i = \frac{\sqrt{3} V_m}{2L\omega} \left[\cos(30 - 30) - \cos(\omega t - 30) \right]$$

$$i = \frac{\sqrt{3} V_m}{2L\omega} \left[1 - \cos(\underbrace{\omega t - 30}_{u}) \right]$$

Given $u = 15^\circ \Rightarrow \omega t = 15 + 30 = 45^\circ$

$$\left.\begin{array}{l} i = 0 \;\; @ \; \omega t = 30^\circ \\ i = I_o \;\; @ \; \omega t = 45^\circ \end{array}\right\} \Rightarrow \quad I_o = \frac{\sqrt{3} V_m}{2L\omega} \left[1 - \cos(u) \right]$$

1d) Average output voltage:

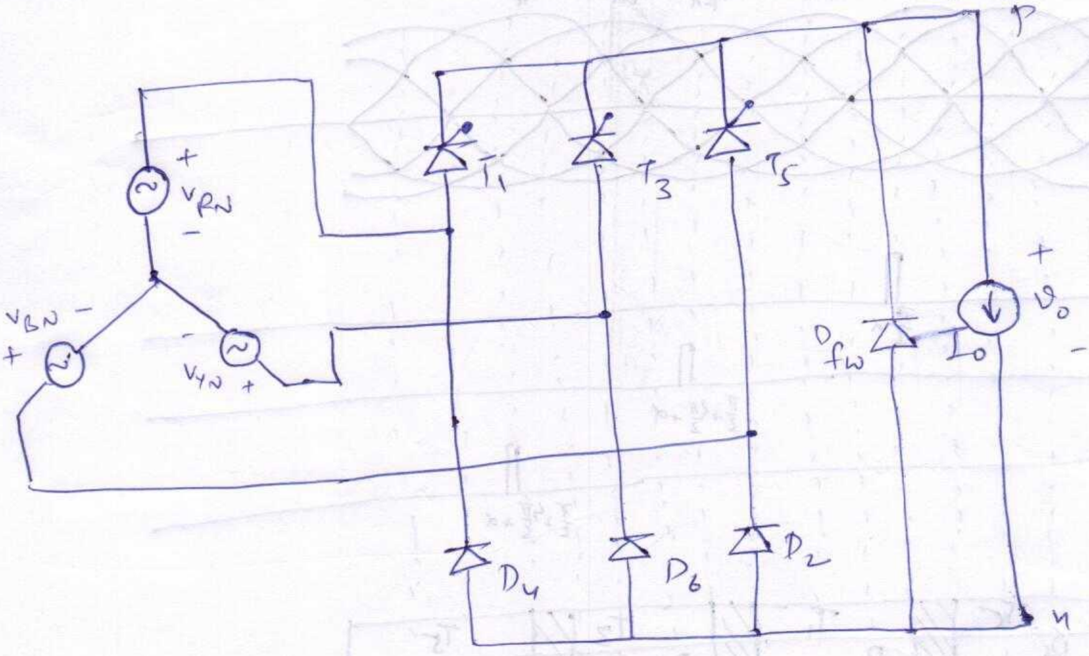
$$\langle V_o \rangle = \frac{3}{2\pi} \left[\int_{\frac{\pi}{6}}^{\frac{\pi}{6}+u} \frac{V_m}{2} \sin(\omega t + 60) \, d\omega t + \int_{\frac{\pi}{6}+u}^{5\pi/6} V_m \sin \omega t \, d\omega t \right]$$

$$= \frac{3V_m}{2\pi} \left[\cancelto{0}{\frac{\cos(30+60)}{2}} - \frac{\cos(30+u+60)}{2} + \cos(30+u) - \cos(150) \right]$$

$$= \frac{3V_m}{2\pi} \left[+ \sin u + \cos(30+u) - \left(-\frac{\sqrt{3}}{2}\right) \right] = \boxed{\frac{3V_m}{2\pi} \left[\frac{\sqrt{3}}{2} + \cos(30-u) \right]}$$

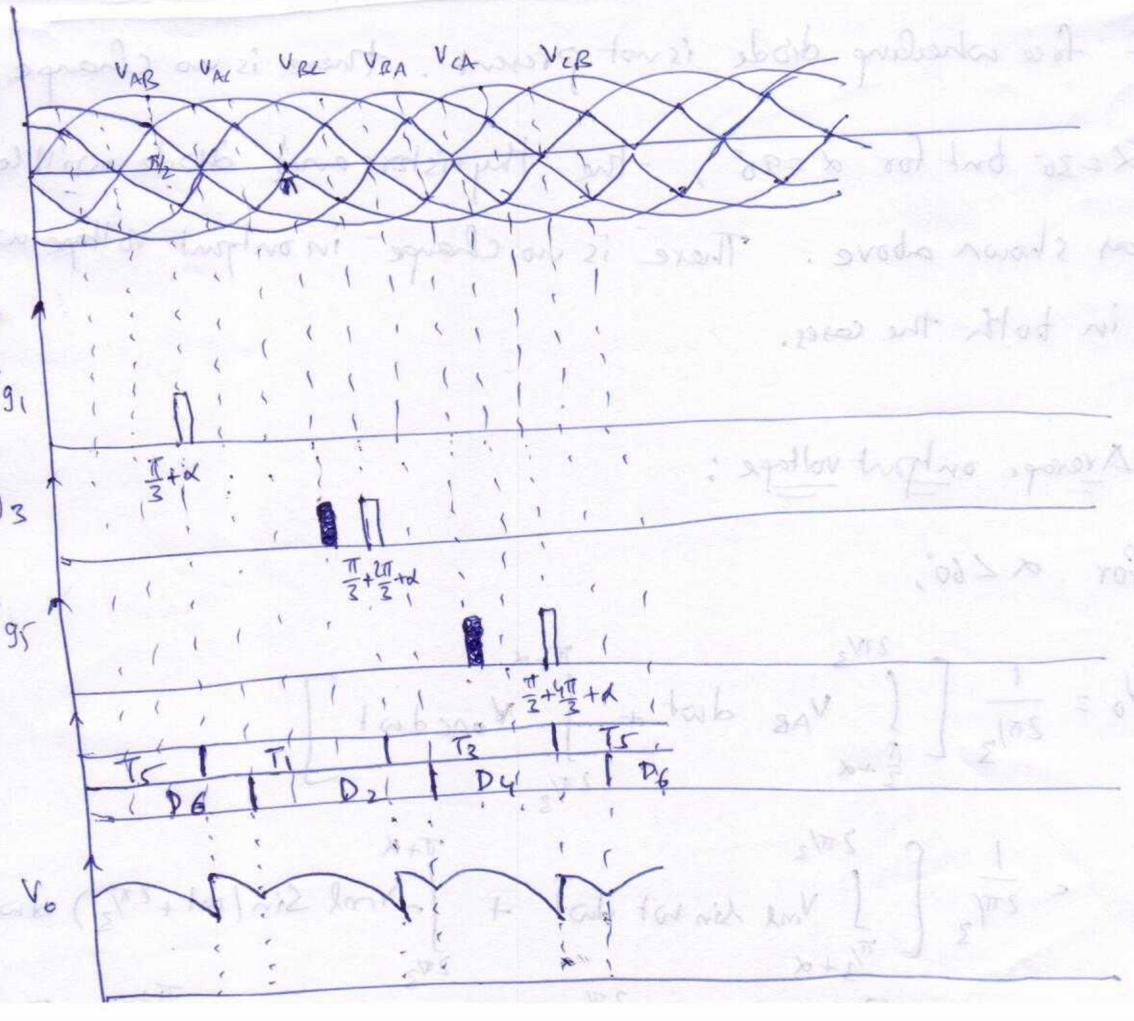
$$u = \cos^{-1} \left[1 - \frac{2 I_o L \omega}{\sqrt{3} V_m} \right]$$

2

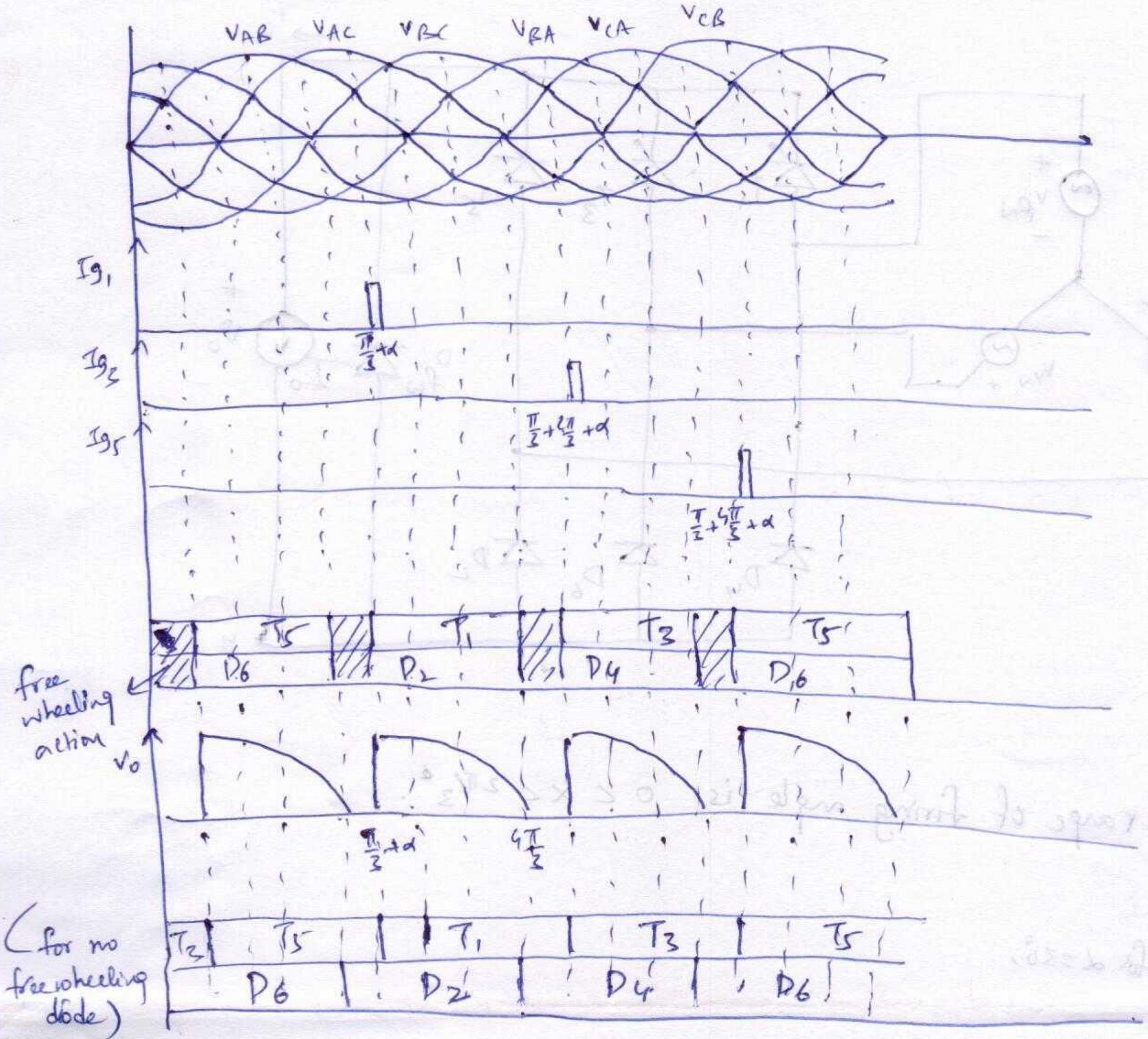


a) The range of firing angle is $0 < \alpha < 2\pi/3$.

b) for $\alpha = 30°$,



$\alpha = 90°$,



If free wheeling diode is not present, there is no change for $\alpha = 30°$ but for $\alpha = 90°$, the thyristors and diode will conduct as shown above. There is no change in output voltage waveform in both the cases.

c) Average output voltage :

for $\alpha < 60°$,

$$V_o = \frac{1}{2\pi/3}\left[\int_{\frac{\pi}{3}+\alpha}^{2\pi/3} v_{AB}\, d\omega t + \int_{2\pi/3}^{\pi+\alpha} v_{AC}\, d\omega t\right]$$

$$= \frac{1}{2\pi/3}\left[\int_{\pi/3+\alpha}^{2\pi/3} V_{ml}\sin\omega t\, d\omega t + \int_{2\pi/3}^{\pi+\alpha} -V_{ml}\sin(\omega t + 2\pi/3)\, d\omega t\right]$$

$$= \frac{3V_{ml}}{2\pi}\left[\cos\left(\frac{\pi}{3}+\alpha\right) - \cos\left(2\pi/3\right) + \cos\left(\frac{2\pi}{3}+\pi+\alpha\right) - \cos\left(\frac{2\pi}{3}+\frac{2\pi}{3}\right)\right]$$

$$= \frac{3V_{ml}}{2\pi}\left[\frac{1}{2}\cos\alpha - \frac{\sqrt{3}}{2}\sin\alpha + \frac{1}{2} + \frac{1}{2}\cos\alpha - \left(-\frac{\sqrt{3}}{2}\sin\alpha\right) - \left(-\frac{1}{2}\right)\right]$$

$$= \frac{3V_{ml}}{2\pi}\left[\cos\alpha - \cancel{\frac{\sqrt{3}}{2}\sin\alpha} + \cancel{\frac{\sqrt{3}}{2}\sin\alpha} + 1\right]$$

$$\boxed{V_o = \frac{3V_{ml}}{2\pi}(1+\cos\alpha)} \qquad (V_{ml} = \sqrt{3}V_{m(ph)})$$

When $\alpha > 60°$,

$$V_o = \frac{1}{2\pi/3}\int_{\pi/3+\alpha}^{4\pi/3} V_{AC}\, d\omega t$$

$$= \frac{1}{2\pi/3}\int_{\pi/3+\alpha}^{4\pi/3} -V_{ml}\sin(\omega t + 2\pi/3)\, d\omega t$$

$$= \frac{3V_{ml}}{2\pi}\Big[\cos(\omega t + 2\pi/3)\Big]_{\pi/3+\alpha}^{4\pi/3}$$

$$= \frac{3V_{ml}}{2\pi}\left[\cos 2\pi - \cos(\pi+\alpha)\right]$$

$$\boxed{V_o = \frac{3V_{ml}}{2\pi}(1+\cos\alpha)} \qquad (V_{ml} = \sqrt{3}V_{m(ph)})$$